Po-Yen (Brian) Lu

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EDUCATION

National Tsing Hua University (NTHU) **Hsinchu, Taiwan**

M.S. Student in Electrical Engineering *Sep. 2023 - Present*

- Advised by Prof. Chao-Tsung Huang

B.S. in Electrical Engineering *Sep. 2019 - Jun. 2023*

- Selected Courses: Introduction to IC Design, Computer Architecture, Advanced Digital Design Flow
- Overall GPA: 4.2/4.3 (Rank: 4/109)

RESEARCH INTEREST

Algorithm exploration, energy-efficient computing architecture design, advanced FinFET digital design flow, digital VLSI system implementation for deep neural networks, computational photography, and generative AI.

PUBLICATIONS

- **Po-Yen Lu**, Po-Wei Chen, Shau-Hua Yeh, Zhi-Jun Zhang, Yong-Tai Chen, Ting-Yu Wang, Chao-Tsung Huang, "Tiamat: A 98-to-134ms/Step Transformer-Based Diffusion Model Processor Supporting Classifier-Free Guidance for Image Generation" in IEEE International Solid-State Circuits Conference (ISSCC), 2026.
- Yong-Tai Chen, Yen-Ting Chiu, Yu-Chou Lee, **Po-Yen Lu**, Mu-Hsien Lee, Jia-Syuan Chen, Kuan-Hsien Ho, Hong-Chuan Liao, Chieh-Cheng Chen, Chao-Tsung Huang, "CATTUS: A 4K-UHD 30FPS Deep Image Processor for Channel Attention Equipped U-Net Acceleration in 16nm FinFET" in IEEE Symposium on VLSI Circuits (VLSIC), 2025.

RESEARCH EXPERIENCE

Vision Circuits and Systems Lab, NTHU **Hsinchu, Taiwan**

Student Researcher | Advisor: Prof. Chao-Tsung Huang *Feb. 2023 - Present*

- Personal High-performance Computing System for Image Generative AI Inference
  - Led a project that developed an energy- and memory-efficient processor for high-quality and real-time image diffusion model inference on edge devices.
  - Proposed a computing architecture optimized for classifier-free guidance (CFG) inference. Devised a memory-efficient inference datapath with microscaling (MX) data format.
  - Tape-out using TSMC 16nm FinFET technology and established a real-time FPGA SoC demonstration platform.
- Deep Learning Accelerator for Image Deblurring
  - Established a complete frontend and backend cell-based digital design flow and hands-on experience in chip tape-out with TSMC 16nm FinFET technology.

- Explored neural network algorithms for single-image super-resolution. Hardware design and implementations for non-linear functions and quantization.

Digital Sound and People Group, NTHU **Hsinchu, Taiwan**

Student Researcher | Advisor: Prof. Yi-Wen Liu *July 2021 - Jun. 2022*

- Detection and Analysis of the Hand Movement in Piano Playing with Triboelectric Nanogenerators
  - Developed a smart glove system for hand movement detection in piano performance, in collaboration with Prof. Zong-Hong Lin on triboelectric nanogenerator integration.
  - Built an initial MIDI-based platform for piano accompaniment.

AWARDS & HONORS

- **Travel Grant**, *Foundation For The Advancement of Outstanding Scholarship (FAOS)* *Jan. 2026*
- **Scholarship**, *Graduate Admission Academic Excellence, NTHU EE* *Oct. 2023*
- **Honorable Mention**, *Integrated Circuit Design Contest* *Aug. 2023*
- **Graduation with Distinction,** *NTHU* *Jun. 2023*
- **Honorable Mention**, *Undergraduate Project Competition, NTHU EE* *Jun. 2022*
- **Scholarship**, *Summer Overseas University Exchange Program, NTHU EE* *Mar. 2022*
- **Academic Excellence Award**, *NTHU* *Fall 2020, Spring 2021, Spring 2022*

TEACHING EXPERIENCE

IC Design Laboratory, NTHU **Hsinchu, Taiwan**

Teaching Assistant (TA) *Sep. 2024 - Jan. 2025*

- Assisted students with lab exercises and assignments, in collaboration with eight other TAs.

Introduction to Semiconductor Design, NTHU **Hsinchu, Taiwan**

Teaching Assistant *Sep. 2024 - Jan. 2025*

- Held office hours and supported students in lecture materials, in collaboration with two other TAs.

Advanced Digital Design Flow, NTHU **Hsinchu, Taiwan**

Teaching Assistant *Feb. - Jun. 2024*

- Co-authored lab documents on low-power design, signoff verification, and logic equivalence check using TSMC ADFP 16nm FinFET technology.
- Assisted students with lab exercises, assignments, and final projects, in collaboration with four other TAs.

SKILLS

- **Spoken Languages**: Chinese (Native), English (Fluent)
- **Programming Languages**: Python, C/C++, Verilog, SystemVerilog, TCL
- **CAD Tools**: Simulator (Cadence NC-Verilog, Synopsys VCS), Lint (Synopsys SpyGlass), Debugging (Synopsys Verdi), Logic Synthesis (Synopsys Design Compiler), Physical Implementation (Cadence Innovus), Sign-off (Synopsys PrimeTime, Cadence Voltus), Equivalence Check (Cadence Conformal)